Algebra Review: Unit 2 — Inequalities

Answer Key

Station 1: One- and Two-Step Inequalities

1. $x > 4$
2. $y \leq 8$
3. $n \geq -5$
4. $m > -4$
5. $x \geq -10$
6. $x < 6$
7. $y \leq -5$
8. $n > 16$
9. $x \geq -4$
10. $25 + 15m \leq 100;\ m \leq 5$

Station 2: Multi-Step Inequalities

1. $x > 6$
2. $x \leq 5$
3. $x > -3$
4. $y \leq -2$
5. $x \geq 5$
6. $x > 9$
7. $x > -3$
8. $x \geq 5$
9. all real numbers
10. no solution

Station 3: Compound Inequalities

1. $-5 < x < 5$
2. $2 \leq x \leq 5$
3. $-1 < x \leq 4$
4. $1 \leq x < 7$
5. $x < 1$ or $x > 7$
6. $x \leq -3$ or $x \geq 3$
7. $-5 < x < -3$
8. $x < 3$ or $x \geq 8$
9. $-2 \leq x < 6$
10. (a) $(-1, 4]$ (b) $(-\infty, 1) \cup (7, \infty)$

Station 4: Absolute Value Equations and Inequalities

1. $x = 7,\ x = -7$
2. $x = 8,\ x = -2$
3. $x = 4,\ x = -5$
4. $x = 6,\ x = -2$ $\quad |x - 2| = 4$
5. no solution *an absolute value is never negative*
6. $-4 < x < 4$
7. $x \leq -4$ or $x \geq 6$
8. $-2 \leq x \leq 5$
9. $x < -5$ or $x > 1$
10. $|L - 5| \leq 0.02;\ 4.98 \leq L \leq 5.02$